



Design Project IOT APPLICATION

SMART IRRIGATION AND SOIL ANALYSIS

SUBMITTED BY-

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SUBMITTED TO -

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PROBLEM STATEMENT

- Lack of real-time soil health data
 - Traditional soil cards are delayed and static
 - Results in inefficient irrigation and crop loss
 - Need for smart advisory and automation system
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PROJECT OBJECTIVES

- Monitor pH, moisture, and temperature in Realtime
 - Automate irrigation when moisture is low
 - Recommend pH correction
 - Suggest crops based on fertility index
 - Provide digital Soil Health Card
 - Send real-time alerts and advice to farmers
- 



SYSTEM ARCHITECTURE

- ESP32 for control and connectivity.
 - Moisture and temperature sensors.
 - Relay and water pump for irrigation.
 - Cloud dashboard for data visualization.
 - Mobile/Web interface for farmer access.
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TECHNOLOGY STACK

- Microcontroller: ESP32
 - Sensors: pH, Moisture, Temperature
 - Cloud: Firebase
 - UI: Blynk
 - Communication: WiFi
 - Power: Battery (optional)
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BRAINSTORMED IDEAS

- IoT soil sensor network with wireless mesh.
 - Integration with local weather APIs for predictive irrigation.
 - AI model that learns seasonal trends per farm.
 - Visual, printable soil health cards with QR access.
 - Farmer 's WhatsApp chatbot integration.
 - Use of voice-based UI for illiterate users.
 - Multi-language app with infographic-based suggestions.
 - Solar-powered system for energy sustainability.
 - Fertilizer dosage calculator based on sensor data.
 - Community soil database for benchmark comparison.
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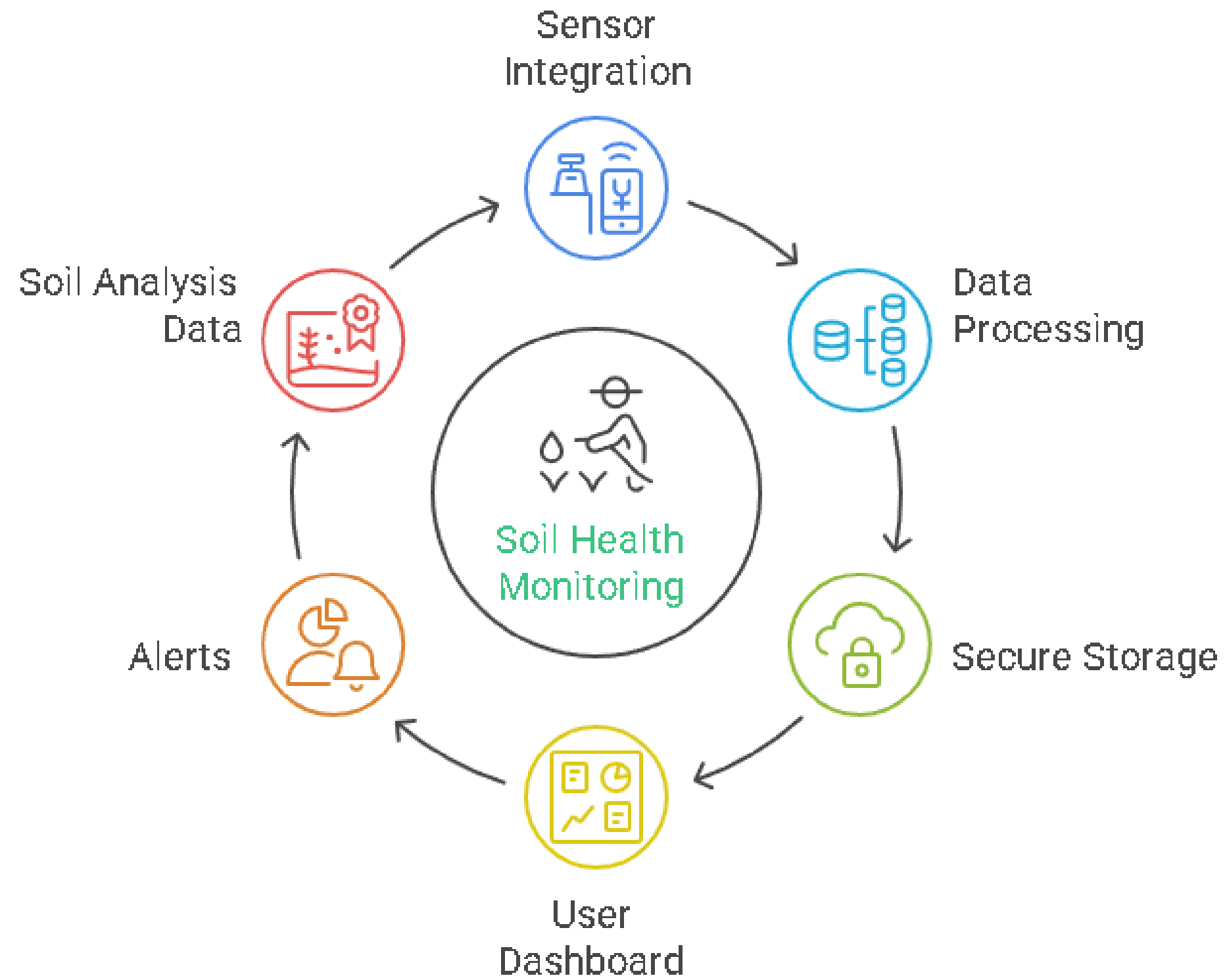
DIFFRENT IDEA

Idea No.	Feature Description
1	AI-Driven Fertility Forecast: Predict crop suitability for next season based on soil trends.
2	Voice Assistant for Farmers: Real-time updates in local language (via speaker or app).
3	QR-Based Soil Card: Printable QR code that shows digital updates when scanned.
4	Weather-Aware Irrigation Logic: Skip irrigation if rain is predicted within 12 hours.
5	Community Benchmarking Dashboard: Compare your soil fertility with nearby farms anonymously.
6	Chatbot on WhatsApp/Telegram: Get soil alerts, advice, and crop calendar via chat.

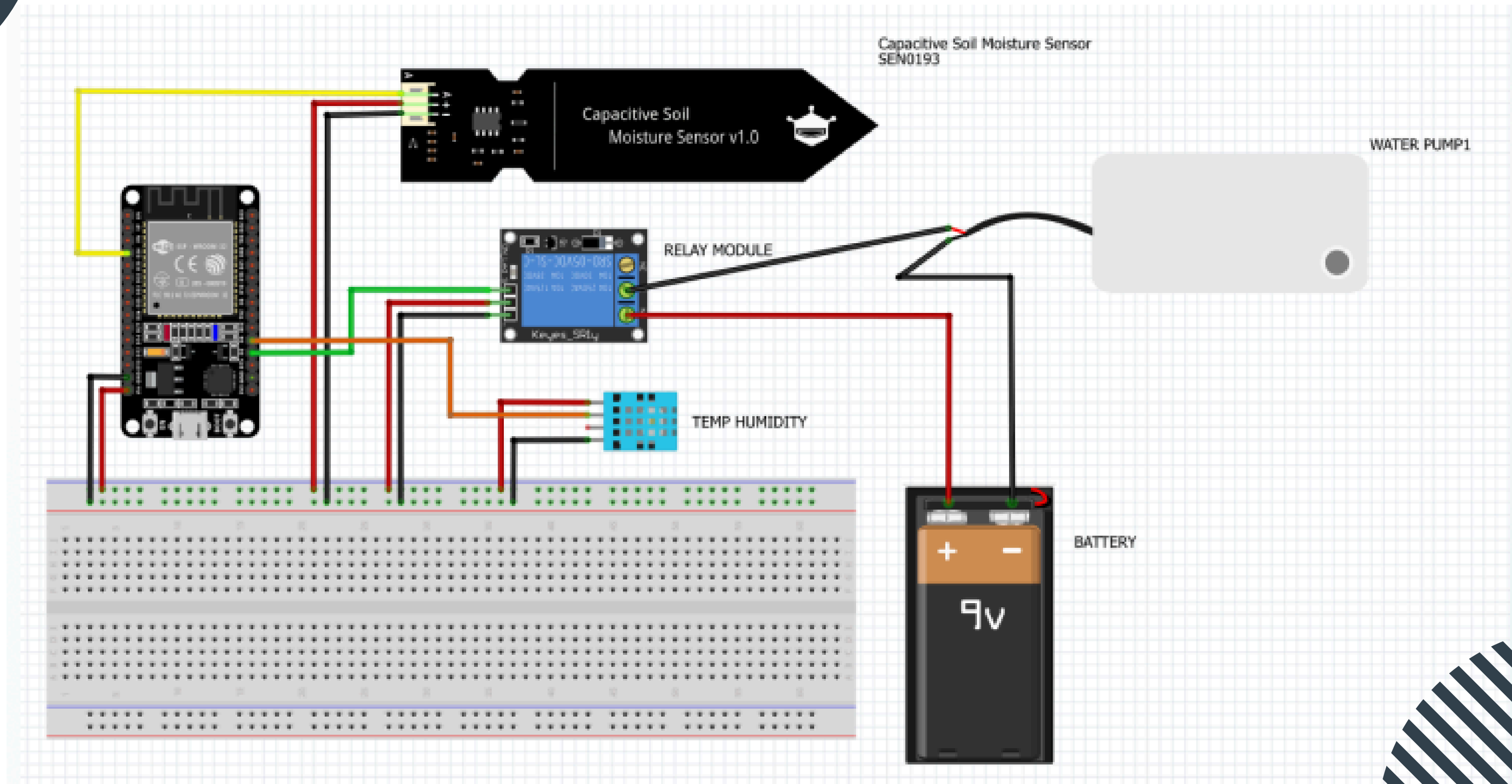
COMPETITIVE ANALYSIS

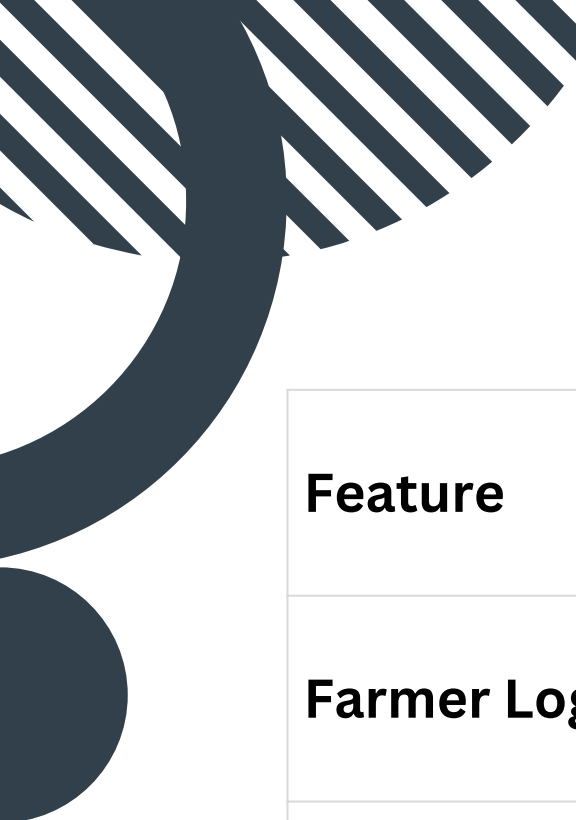
Feature	Your Solution	Kisan Suvidha App	Krishi Network	mKisan Portal
Real-Time Soil Monitoring	Yes	No	No	No
Automated Irrigation	Yes	No	No	No
pH Correction Advisory	Yes	No	No	No
Fertility Index & Crop Recs	Yes	No	Partial	Partial
Mobile/Web UI	Yes	Yes	Yes	Yes
Alerts & Notifications	Yes	Yes	Yes	Yes
IoT Integration	Yes	No	No	No
Multilingual & Accessible	Planned	Yes	Yes	Yes

THOUGHT PROCESS



CIRCUIT DIGRAM





TESTING CHART

Feature	Input	Expected Output	Result
Farmer Login	Valid username/password	Login successful	PASSED
Farmer Login	Invalid credentials	Error message: "Invalid login"	PASSED
Crop Selection	Select "Wheat"	Ideal ranges loaded for wheat	PASSED
Moisture Check	Moisture < crop threshold	Pump ON for that area	PASSED
Manual Pump Control	Farmer clicks ON for Area 2	Pump starts for Area 2	PASSED
Map Display	Sensor in Area 3 disconnected	Dashboard shows "Sensor Offline"	PASSED
Multi-Area Data	Different readings per area	Dashboard displays separate values	PASSED



TESTING

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PLANT

Smart Irrigation System

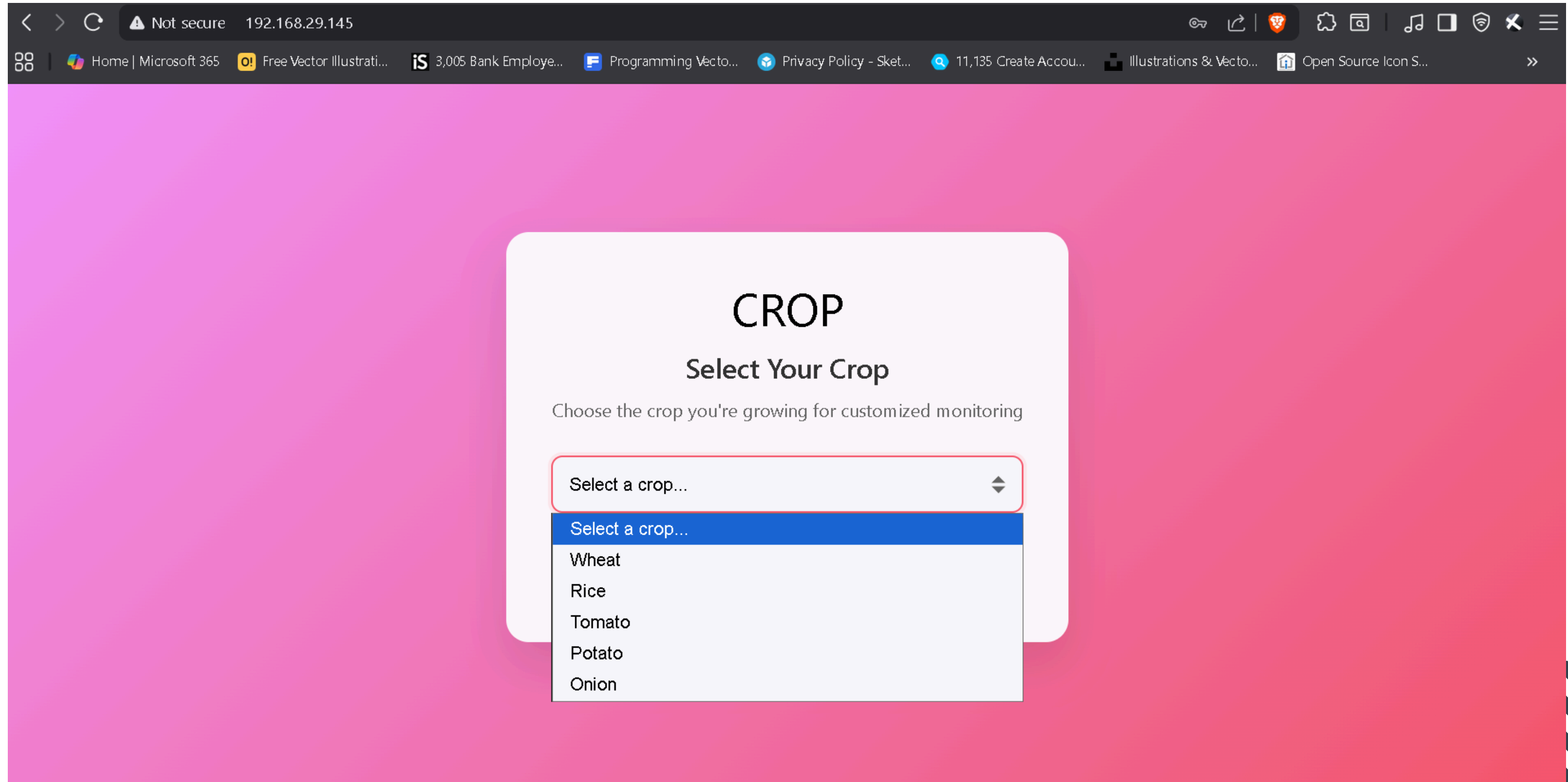
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Password

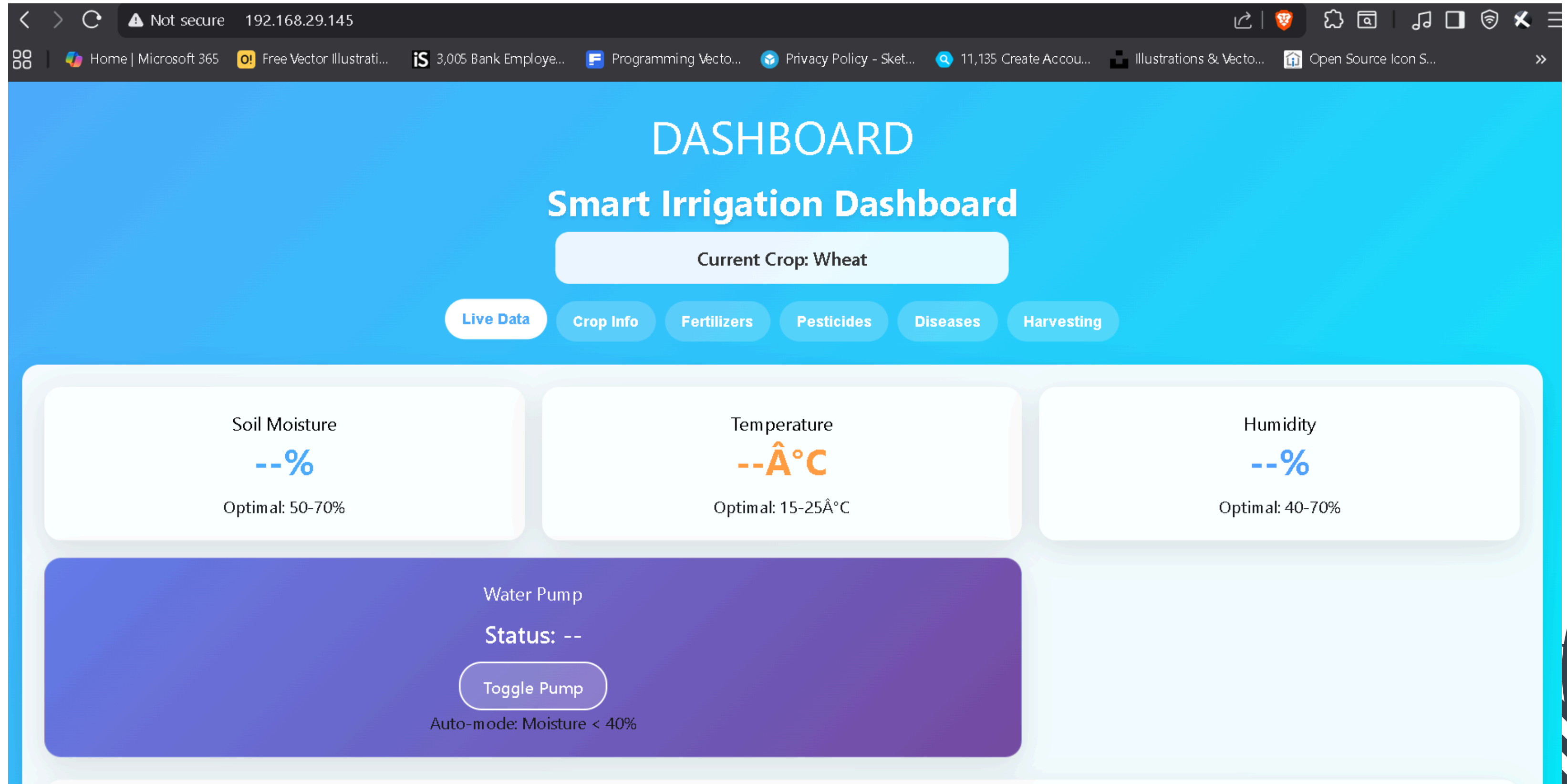
Login to Dashboard

Secure Agricultural Monitoring

TESTING



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DASHBOARD

Smart Irrigation Dashboard

Current Crop: Wheat

Live DataCrop InfoFertilizersPesticidesDiseasesHarvesting

About Wheat

Description: Wheat is a cereal grain grown for its seed, a staple food used to make flour, bread, pasta, pastry, etc.

Growing Season

Rabi (Winter) Season - Oct to Mar

Growth Period

110-130 days

Water Requirements

Moderate - 450-650 mm per cycle

Pro Tips

Rotate crops to prevent disease, Use certified seeds, Monitor for aphids

Change Crop

Logout

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DASHBOARD

Smart Irrigation Dashboard

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Crop Info

Fertilizers

Pesticides

Diseases

Harvesting

Harvesting Guide for Wheat

Harvesting Indicators

When grains are hard and moisture content is 20-25%

Best Time

Harvest in morning hours, avoid rainy days, use proper tools

Post-Harvest

Proper cleaning, grading, storage in cool dry place, monitor for storage pests

Change Crop

Logout

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DASHBOARD

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Diseases

Harvesting

Common Diseases in Wheat

Major Diseases
Rust, Smut, Karnal Bunt, Powdery Mildew

Prevention
Use disease-free seeds, practice proper spacing, avoid waterlogging

Treatment
Apply fungicides at first symptoms, remove infected plants, improve air circulation

Change Crop

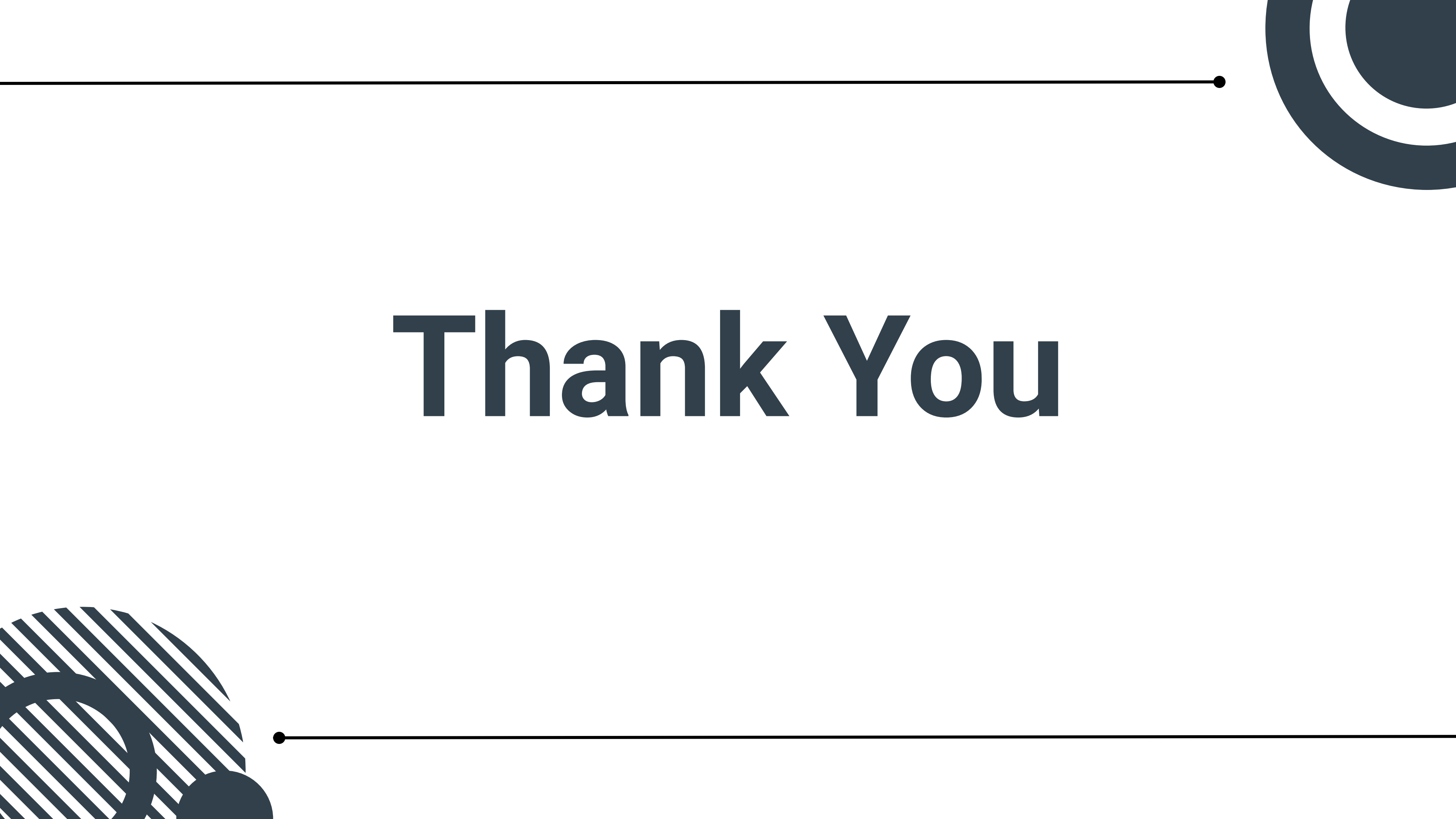
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FUTURE ENHANCEMENT

In the future, the system can integrate weather control mechanisms alongside soil monitoring, including

- Automatic or Manual Land Cover Control (Motorized Shed System)
 1. In heavy rainfall: The shed covers the land to prevent crop damage and channels rainwater into a storage tank. Stored water can later be used for irrigation.
 2. In extreme summer heat: The shed covers the land to protect crops from excessive sunlight and reduce evaporation losses.
 - Dashboard Control & Alerts
 3. The farmer can manually open/close the shed from the mobile app or web dashboard.
 4. Live weather data and shed status are shown in the app.
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Thank You
